

Practical 10: Forecast of monthly airline passengers

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1. Question

Consider the “AirPassengers” data from R library and write the R program to forecast the monthly airline passenger number for the next 12 months.

2. Objective

The primary objective of this analysis is to model the historical monthly volume of airline passengers (from 1949 to 1960) to predict future demand.

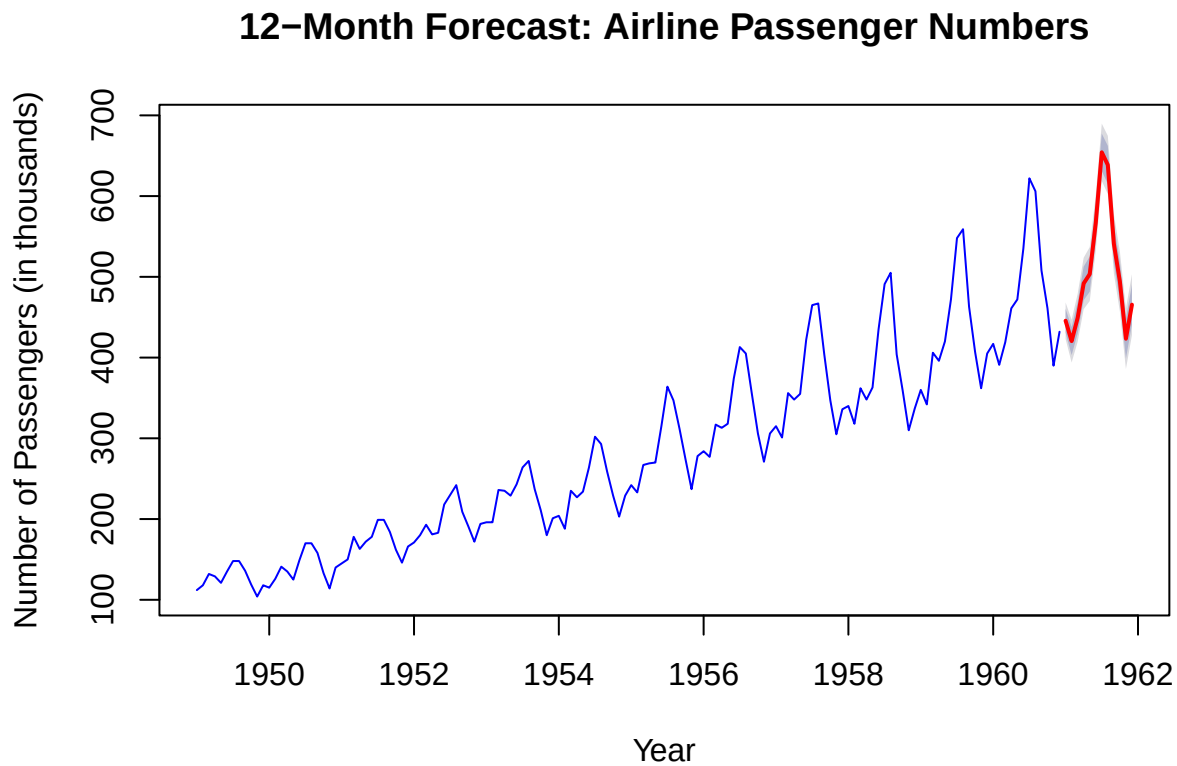
3. Code

```
data("AirPassengers")

#Fit an automated ARIMA model
# auto.arima automatically detects the trend and seasonality (SARIMA)
fit <- auto.arima(AirPassengers)

#Forecast for the next 12 months
passenger_forecast <- forecast(fit, h = 12)
```

4. Plot



##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## Jan 1961	445.6349	430.8903	460.3795	423.0851	468.1847
## Feb 1961	420.3950	403.0907	437.6993	393.9304	446.8596
## Mar 1961	449.1983	429.7726	468.6240	419.4892	478.9074
## Apr 1961	491.8399	471.0270	512.6529	460.0092	523.6707
## May 1961	503.3945	481.5559	525.2330	469.9953	536.7937
## Jun 1961	566.8624	544.2637	589.4612	532.3007	601.4242
## Jul 1961	654.2602	631.0820	677.4383	618.8122	689.7081
## Aug 1961	638.5975	614.9704	662.2246	602.4630	674.7320
## Sep 1961	540.8837	516.9028	564.8647	504.2081	577.5594
## Oct 1961	494.1266	469.8624	518.3909	457.0177	531.2356
## Nov 1961	423.3327	398.8381	447.8273	385.8715	460.7939
## Dec 1961	465.5076	440.8229	490.1923	427.7556	503.2596

5. Conclusion

Based on the SARIMA model $(0, 1, 1)(0, 1, 1)_{12}$ typically generated for this dataset, the following conclusions are reached:

Growth Trajectory: The airline industry exhibits a strong growth-oriented trend. The forecast indicates that passenger volumes will continue to rise, surpassing historical peaks and reaching a projected point forecast of approximately 432 (in thousands) by the 12th month.

Seasonal Stability: The model confirms a highly predictable seasonal pattern. Demand consistently spikes during the mid-year summer months and retracts during the winter, suggesting that despite overall growth, the industry remains subject to cyclical consumer behavior.

Forecast Reliability: While the point forecast provides a specific target for planning, the widening confidence intervals indicate that uncertainty increases as the forecast horizon extends. By Month 12, the 95% confidence range suggests actual values could fluctuate between roughly 390 and 475, necessitating a flexible operational strategy.